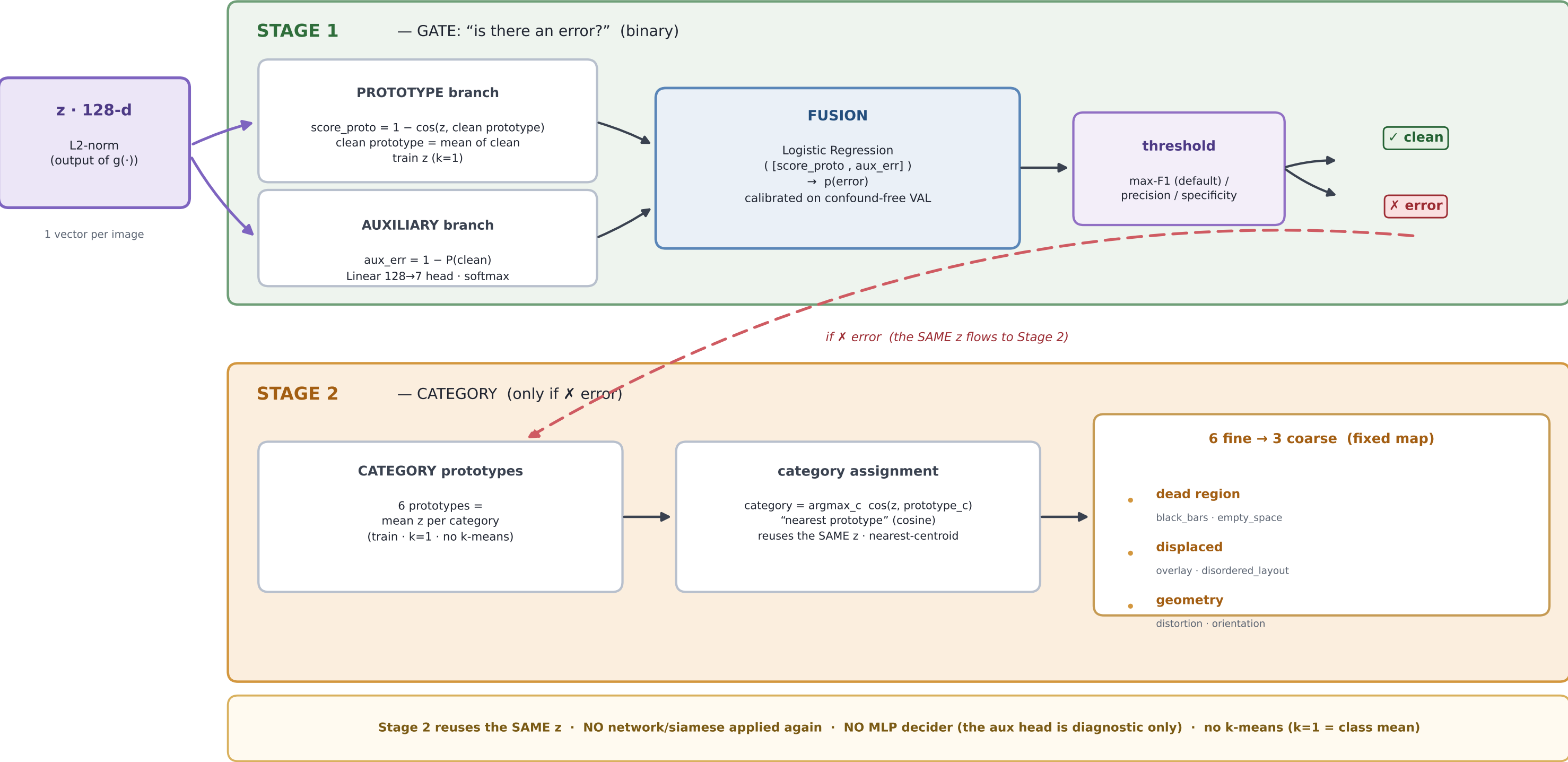


Two-stage decision

both stages reuse the SAME z vector (128-d, L2-norm) already produced by the projection head $g(\cdot)$ — no network is applied again



training (offline): $g(\cdot)$ shapes z via SupCon + 0.6·CE; prototypes = class means in z . inference: 1 dot product + arg max — no trained weights in Stage 2.